

AKASH SINGH

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EDUCATION

Indian Institute of Information technology

Bachelor of Technology Computer Science

Bhopal, Madhya Pradesh

August 2023 - July 2027

Relevant Coursework: Data Structures and Algorithms, Object Oriented

Programming, Database Management Systems, Operating Systems, Computer Networks

EXPERIENCE

NirveonX Omnicare Pvt. Ltd.

AI Backend Engineer Intern

Hyderabad, Remote

January 2026 – March 2026

- Built and shipped scalable backend services using Python, FastAPI, and PostgreSQL, enabling real-time AI workflows with optimized request-response handling and reduced latency in production systems.
- Debugged complex production issues, performed root-cause analysis, and implemented performance optimizations across distributed systems, ensuring high reliability and stable deployments.
- Designed modular AI service layers integrating LLM APIs (OpenAI/Gemini), Redis caching, and asynchronous workflows, improving backend scalability and enabling execution of dynamic AI pipelines.

CollegePur

Backend Developer Intern

Noida, Remote

November 2025 – December 2026

- Developed and shipped backend APIs using Python and REST architectures, ensuring reliable communication across services and seamless integration with frontend application layers.
- Identified and resolved integration and data consistency issues across services and databases, improving system reliability and ensuring accurate data synchronization across backend modules.

Eastri

Software Developer Intern

Mumbai, Remote

July 2025 – september 2025

- Integrated third-party and internal APIs using scalable backend systems, enabling efficient data pipelines and reliable communication across application services.
- Diagnosed and resolved issues across frontend and backend systems, improving application stability, reducing error rates, and ensuring smoother production release cycles.

PROJECTS

AI SaaS Platform – Full Stack AI Application | .js, TypeScript, Python (FastAPI), PostgreSQL, LLM API(Gemini), LangChain, Redis [Github](#)

- Built and deployed a scalable full-stack AI application using Next.js, FastAPI, PostgreSQL, and LLM APIs (Gemini), enabling real-time user interaction and dynamic content generation.
- Designed modular backend pipelines with LangChain-style orchestration and structured prompt workflows, ensuring maintainability and scalable AI service integration.

TravelAI – Smart Group Travel Planning Platform | Next.js, React, TypeScript, Python (FastAPI), PostgreSQL, Pinecone, FAISS, Hybrid Search, LLM APIs [Github](#)

- Built an AI-powered travel planning platform using LLMs and RAG pipelines, generating personalized itineraries based on user preferences and contextual constraints.
- Engineered a semantic retrieval system using Pinecone, embeddings, and hybrid search (BM25), improving recommendation accuracy and retrieval efficiency.
- Designed evaluation datasets and benchmarking pipelines using Recall@K and A/B testing, optimizing personalization quality and system performance under load.

PayFlux AI – Smart Payment Gateway | .js, Express, React Native, MongoDB, REST APIs, Distributed Systems Design [Github](#)

- Implemented intelligent transaction validation pipelines with rule-based scoring and real-time classification logic, ensuring fault-tolerant execution under unreliable network conditions.
- Designed and developed an offline-first payment system using Node.js, MongoDB, and distributed backend workflows, enabling queued transactions during network failures.
- Built scalable backend workflows ensuring transaction consistency, retry mechanisms, and reliable execution across distributed systems.

SKILLS

AI Systems & GenAI	LLM APIs (OpenAI, Google), RAG Pipelines, AI Agents, Prompt Engineering
Backend & Distributed Systems	Python (FastAPI, Django), Go (concurrent services), REST APIs, Microservices
Full Stack Development	Next.js, React, TypeScript, Node.js, Production UI + API Integration
Data & Retrieval Systems	FAISS, Pinecone, Hybrid Search (BM25 + Embeddings), Reranking Pipelines
Infrastructure & Streaming	Redis, Kafka, Event-driven Architecture, Docker, GitHub Actions
Evaluation & Languages	Recall@K, A/B Testing, Python, Go, JavaScript, Java, C/C++

ACHIEVEMENTS

- Attained a top 10 position in the Algo university contest among the 1000 participants.
- Secured a top rank in inter-college coding competitions, consistently solving complex problems.